



PROSTHETIC ARM USING EMG SENSOR

MINI PROJECT REPORT

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ADITYA U TAWKER

ARAVIND LN

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(An Autonomous Institution; Affiliated to Anna University, Chennai -600 025)
ANNA UNIVERSITY :: CHENNAI 600 025

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SRI VENKATESWARA COLLEGE OF ENGINEERING
(An Autonomous Institution; Affiliated to Anna University, Chennai-600025)

ANNA UNIVERSITY :: CHENNAI 600 025

BONAFIDE CERTIFICATE

Certified that this project report “**PROSTHETIC ARM USING EMG SENSOR**”
is the bonafide work of “**ADITYA U TAWKER (190701002),**
ARAVIND LN(190701009)” who carried out the project work under my
supervision.

SIGNATURE

Dr. S. MUTHUKUMAR, M.E., Ph.D

HEAD OF THE DEPARTMENT

Department of Electronics and
Communication Engineering

SIGNATURE

Dr. G.A. SATHISH KUMAR, M.E., Ph.D

FACULTY IN CHARGE
ASSISTANT HEAD OF THE DEPARTMENT

Department of Electronics and
Communication Engineering

Submitted for the project viva-voce examination held on

INTERNAL EXAMINER

EXTERNAL EXAMINER

ABSTRACT

The prosthetic hand is a device used for the amputee to make them capable of doing daily life stuff. Losing upper limbs is not just a physical loss it also affects the way that person sees their life, they start feeling disabled all the time. This event grows and make them disabled mentally or by god if they can handle it even though people will remind them time to time about their disabilities which may lead to suicide or even worse. To join hand to hand and to make more abled than disabled, we are introducing prosthetic hands at the cheapest price. It's not that this event is the only device in the market but I could say this event is the cheapest one with these many qualities. It could not be very good looking but it could be better than a decent one. Disability does not differentiate by the salary of a person or capability of someone, so this should be affordable to most of them. In the west, we have some countries who can make prosthetics at an affordable price but their money and employment are stronger than us. So, a project like this can make a difference. The concept partially fulfilled the goals and targets set at the outset of the project. The initiative originally had four primary goals. The first goal is to create a tool like a prosthetic hand that can help an amputee with hand loss accomplish simple everyday tasks. Because of all the EMG signal calculation of various hand movements performed on a person who does not lose his her hand the system can only partially accomplish this goal.

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ADITYA U TAWKER

ARAVIND LN

TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
1.	INTRODUCTION	
	1.1 INTRODUCTION	1
	1.2 OVERVIEW	2
	1.3 THESIS OBJECTIVE	3
2.	LITERATURE REVIEW	5
	2.1 REMARKS	7
3.	PROPOSED WORK	
	3.1 PROPOSED ARCHITECTURE DIAGRAM	8
	3.2 DESCRIPTION OF THE ARCHITECTURE DIAGRAM	
	3.2.1 DEFINING THE PROBLEM STATEMENT	9
	3.2.2 PROPOSED COST-EFFECTIVE PROSTHETIC ARM	10

4.	SYSTEM REQUIREMENTS	
	4.1 HARDWARE REQUIREMENTS	13
	4.2 SOFTWARE REQUIREMENTS	13
	4.3 DESCRIPTION OF THE TOOLS USED	
	4.3.1 EMG SENSOR	14
	4.3.2 EMG CABLES WITH ELECTRODES	15
	4.3.3 ARDUINO BOARD	16
	4.3.4 SERVO MOTOR	17
	4.3.5 PROSTHETIC ARM	18
5.	EMG SIGNAL PROCESSING	
	5.1 INTRODUCTION	19
	5.2 EMG SIGNAL DETECTION AND PROCESSING	19
6.	PROGRAMMING THE ARDUINO	21
7	CONCLUSION AND FUTURE WORK	
	7.1 CONCLUSION	23
	7.2 FUTURE WORK AND RECOMMENDATION	25
	REFERENCES	26

LIST OF FIGURES

FIGURE NUMBER	NAME OF THE FIGURE	PAGE NUMBER
3.1	PROPOSED ARCHITECTURE DIAGRAM	8
4.3.1	EMG SENSOR	15
4.3.2	ELECTRODES	16
4.3.3	ARDUINO MEGA	16
4.3.4	SERVO MOTOR	17
4.3.5	PROSTHETIC ARM	18
5.2	EMG SIGNAL DETECTING AND PROCESSING	20

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The prosthetic hand is an artificial device which is designed for the people with upper limb amputations to come up with some functions of the natural hands.

Loss of upper limbs has many circumstances not only in the physical sense of physically but also in the social and psychological department. There are many commercial prosthetic hands available these days in the market. One of its main goal of our project is to develop a lower-cost 3D prosthetic hand printed for amputees. A standard myoelectric hand, where all the EMG surface electrodes on the amputee hand capture myoelectric signals that are manipulated to enable the hand to open and close. In the past decade, advances in 3D printing have made 3D trans-radial prosthetics a cheap substitute. It provides robust features at an inexpensive price. There are many designs these days that vary in capacity and operate quite differently, such as reliability, control mechanism, hand grips, user interfaces, convenience and cost. Every prosthetic hand has strengths and weaknesses of its own. The main aim of this bachelor's thesis is to create a 3D printed prosthetic hand that can be operated from the forearm using an EMG signal. The prosthetic to replace the missing hand must be attached to the patient's forearm. To monitor the impulse, three electrodes (EMG stickers) that read the EMG signal will be attached in the forearm at different locations.

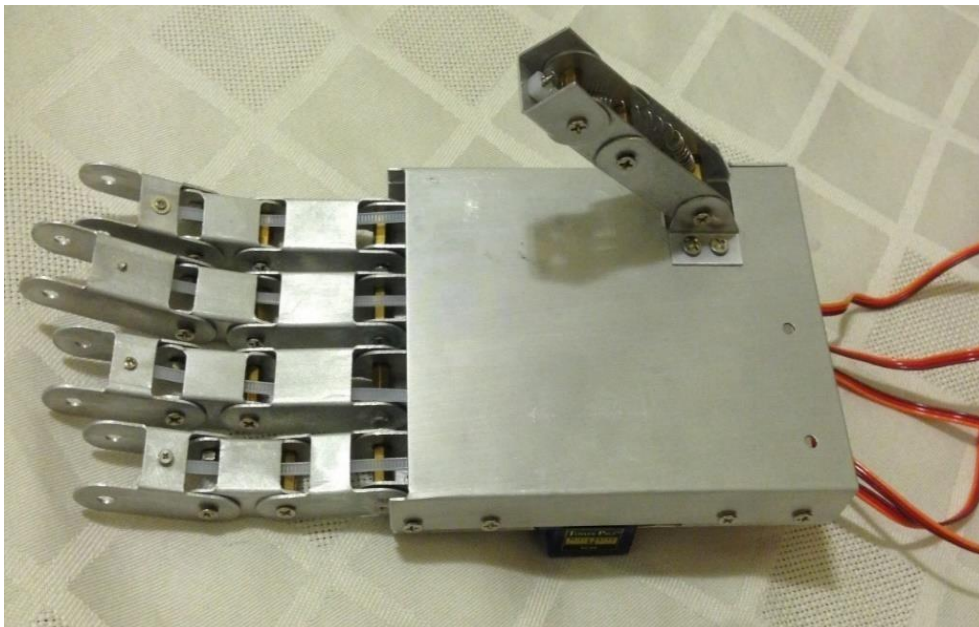
1.2 OVERVIEW

The primary purpose is to replace the function of a missing limb. Prosthetic arm turns the human beings with limb difference into independence with its robotic hand and makes science fiction a reality. Recognizing the need for an alternative to currently available technology. For those with resources, we are able to develop a prototype hand similar functionally to the more sophisticated myoelectric hands on the market. While reducing the cost of the hand to well below that of the most affordable Myoelectric hands that are currently available. The main objective is to achieve a hand by developing an inexpensive EMG control platform and a mechanical hand that is designed to be produced inexpensively on a 3D Printer. Our goal of achieving a high level of hand functionality is achieved by implementing a user interface that allows many more hand functions to be called by the user than current myoelectric hand control systems.



1.3 THESIS OBJECTIVE

This thesis is motivated by the challenge to develop a prosthetic hand that is not heavy, it is very cost-effective (to be affordable by almost every amputee), with effective and efficient feedback sensing mechanisms and capable of reproducing most of a human hand's movement and gripping patterns. The prosthetic hand should be strong enough to hold different object shapes and weight, and its size should be comparable to a human hand. As such, the thesis focuses on designing a cost-effective five-finger prosthetic hand. Aluminum is chosen as the building material. Each finger is actuated separately by controlling the angle of a corresponding servo motor. The five fingers combined, account for 14 DOF. Each finger except the thumb, has three links, one base and three joints. The thumb has two links, one base and two joints.



Each finger uses a force sensor to sense objects (in touch with the finger) and returns a feedback signal. The prosthetic hand is controlled by surface EMG signals obtained from surface electrodes that are connected to the arm muscle(s). Obtained signals from the electrodes go through a controller to generate specific finger/hand motions, and also provide the necessary feedback. For the proposed design, five muscles are used to reproduce hand motions.

The most challenging task of designing the prosthetic hand is EMG signal detection and processing. It starts with connecting (three) multi-useable electrodes, positive, negative and ground, to the skin/surface covering the muscle, in order to detect muscle movement.

The ground electrode is connected close to any bone; the positive/negative electrodes of each muscle are connected to the top and bottom end or to the middle of the muscle.

The EMG signal passing through the electrode wires may be affected by electromagnetic fields close to the wires [3], and this may generate undesirable noise in the EMG signal. Twisting the electrode wires (see Figure 3.1b) prevents interference and reduces noise as electromagnetic fields in opposite directions cancel each other. After EMG signal detection, EMG signal processing follows, which includes differential, pre- amplifier, filtering and amplifier stages. The five fingers are controlled by an Arduino UNO depending on the received EMG signal(s). A force sensor is attached to each fingertip to sense any object in touch with a finger. These force sensors turn on and off vibration motors connected to the amputee's arm, informing the amputee if there is anything in touch with the fingertip. Note that the proposed EMG signal detection, processing and control approach is dependable and reliable, accurate, repeatable and cost-effective, while the corresponding circuit size is really small.

CHAPTER 2

LITERATURE REVIEW

S. K. Choudhary, D. Chakraborty, N. M. Kakoty, and S. M. Hazarika, “Development of cost effective EMG controlled three fingered robotic hand,”

The authors present the development of a low cost three finger robotic hand with six degrees of freedom. The hand has three actuators and it can grasp different shapes such as oval, cuboid, circular and cylindrical. The grasping command depends on the EMG signals coming from three surface electrodes.

The raw EMG signals from electrodes go to the EMG preprocessing unit.

This unit consists of a differential amplifier, which subtracts the two signals that come from the muscle; the first signal comes from the electrode that is connected to the middle of the muscle and the second signal comes from the electrode connected to the end of muscle. This step is used to ‘stabilize’ the signal.

There is a pulley with tendons wrapped around it. In order to transmit the force from the Servo motor to the pulley, a nylon thread is used to extend and flex the links of each finger. There are limitations to this design. The first limitation is that there are just two links in each finger so the fingers cannot reach very many points in the working area.

P. Geethanjali and K. K. Ray “A Low-Cost Real-Time Research Platform for EMG Pattern Recognition-Based Prosthetic Hand,” A surface EMG signal is used to allow for the prosthetic hand the ability to evaluate six different hand motions. There are three electrodes connected to each muscle, two electrodes which measure the differential voltage in two different points in the muscle and the third electrode that serves as ground. Fingers have been designed using CAD software. Components were built by using rapid prototyping techniques with polypropylene. Each finger has two revolute hinge joints and three degrees of freedom. The reason behind having this simple structure is that abduction/adduction movements of the metacarpophalangeal joint have been ignored. There are holes made on phalanges to route the drive string through all the phalanges. Using the Servo motor, nut, and bolt mechanisms, the string can be pulled/ pushed so the finger will be closed or opened, depending on the motor’s direction of rotation, which is decided by the order coming from the EMG’s signals

C. Scott, G. Sen Gupta, and L. Tang, “Sensing and processing of biometric signals for use in low cost bio-robotic systems,” The goal is to develop a cost-effective robotic hand controlled by EMG signals collected from surface skin electrodes. There are two ways to obtain signals from muscles. Each method has advantages and disadvantages. The invasive way allows for very readable signals but, sometimes, there may be some side effects, like infection or body rejection to electrodes. The hand’s design consists of motorized digits, which are either extended or contracted depending on the EMG signal that comes from the muscle. Limitations of this design are: there are no feedback sensors; the used filter does not account for the signals with 50 HZ (which is the electricity waves).

2.1 Remarks

Looking at the literature, it is obvious that there is a long list of design considerations to design, develop and control a cost-effective EMG prosthetic hand. In this thesis, priority is given to size, weight, complexity, cost effectiveness, as well as efficient functionality. Many of previous designs address some of these attributes, but it appears that there is no cost-effective design that addresses all of them, in unison, for a single prosthetic arm – and this has motivated the proposed design.

CHAPTER 3

PROPOSED WORK

3.1 PROPOSED ARCHITECTURE DIAGRAM

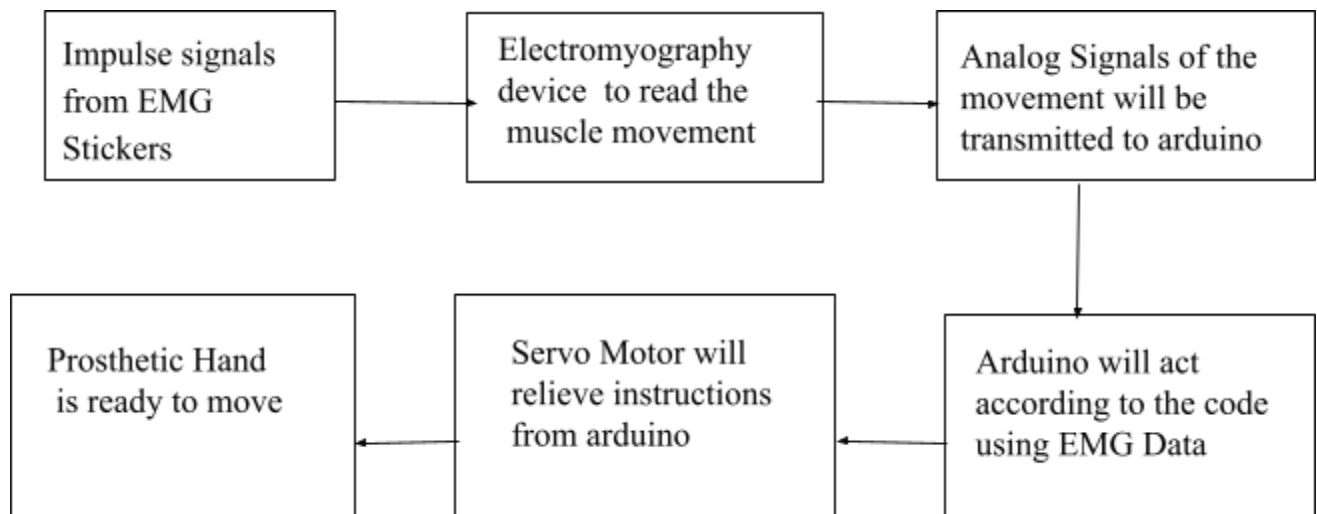


Figure 3.1 Proposed Architecture Diagram

3.2 DESCRIPTION OF THE ARCHITECTURE DIAGRAM

3.2.1 Defining the Problem Statement

The human body is amazing. Each part of the body has a specific function, yet when combined with all the other parts working together, the body is a magnificent machine. Just as with other machines, when parts of the machine/body break down, some human parts can be transferred from one person to another; some human parts can be replaced with man-made parts; and when some human parts no longer work, and the body cannot be ‘repaired’, it will stop working altogether. One of the most important parts of the human body is the hand. The hand movements help the body perform such actions as pushing, pulling and/or grasping objects; feeding one’s self, writing, etc.

The hand is structured to handle objects of different sizes, shapes, and weight. It is covered by skin with nerve endings to sense the external world through touch. Use of the arms and hands assists in balancing the person’s body in space, either while being still, as on a tightrope, or while moving, as in running.

Living without a hand or hands becomes very challenging for the amputee. Many attempts have been made towards resolving problems for amputees to help them live a more normal life. History reports that an early surgery, done between 950 – 710 B.C., discovered on a mummified Egyptian corpse, attempted to create a solution for the loss of a big toe by using a piece of wood to replace a big toe.

The first documentation of the use of an artificial hand occurred between 218-201 B.C., for Marcus Sergius, a Roman general who lost his right hand during a battle. Sergius used an iron hand to hold his shield and went back to the battle to fight again. Since then, the field of creating / developing artificial hands has continued to grow. The first study of EMG was done by Francesco Redi in 1666, followed by the first record of EMG signals by Jules Marey in 1890. Marey was also the first person to use the term “electromyography”. Great progress in the field of artificial hands occurred when the EMG signal was discovered to be able to control the prosthetic hand. Researchers continued to try to design a prosthetic hand which could satisfy the needs of the amputees. The best designs found to be useful to amputees are those that can mimic the actual movements of the human hand, have good sensory feedback, and create enough force to handle objects with smooth movements and have very good EMG signal detection and processing. A prosthetic hand that includes all these requirements in one prosthetic device would make it cost prohibitive for most amputees. This research has focused on the creation of an affordable prosthetic hand that functions ‘well enough’ to make it available and useful for amputees.

3.2.2 PROPOSED COST- EFFECTIVE PROSTHETIC HAND

Creating an inexpensive prosthetic hand requires many steps, beginning with a way for the prosthesis to detect the EMG signal. A major concern was to determine which kind of electrodes should be used. If regular electrodes were used, this would require changing the electrodes frequently, which would increase cost over time. Instead, it was determined and decided to use multiuse electrodes for / during the EMG detection process.

The next decision was to determine the sequence of steps (process) to best transfer the detected signal(s) from the electrodes without ‘transferring’ the noise, which results from the electromagnetic field coming from other nearby electrical devices. There are two ways to minimize such noise. The first one is to use active electrodes [20], i.e., use very short wires to connect the electrodes with the circuit, as shown in Figure 3.1a. The second is to twist the wires that transfer the signal, as shown in Figure 3.1b. By twisting the wires, the space between the wires is divided into many small areas. In this way, the electromagnetic field resulting from one small area is canceled by the electromagnetic field that results from the next area. Both of these methods can be used, depending upon the situation and the length of the resting arm. Sometimes long twisted wires are needed to connect the electrodes, and sometimes the use of active electrodes is better. A high-quality, low-price EMG Sensor will be used to subtract the two incoming signals from the two electrodes that connect to the muscle. This subtraction process will eliminate the noise that results from reading signals from more than one muscle. The pre-amplifier stage will begin to make the detecting signal large enough to process by the use of low and high pass active filters. This filtered signal is amplified to be readable by the Arduino.

This circuit reads each muscle’s activities as an electrical signal and sends it to the controller to create an action. This circuit has a lightweight, low price, and small size, and produces reliable results. Therefore, it can be used in many applications, such as to aid with a prosthetic limb or other EMG controlled projects. After reading the EMG signal by the Arduino, the Arduino makes a decision to control the finger or fingers of the hand, depending upon the EMG signal. There are many functions in the Arduino, which make it perfect to use.

The Arduino has six analog pins (from A0 to A5) that can be used for inputting or outputting analog signals.

Once the EMG signal is completed, the signal becomes readable by the Arduino, and the Arduino may then control the prosthetic hand. The biggest challenge now is to design a lightweight, human-sized hand, with good feedback and strong enough to complete the activities of the human hand. One design could be completed by using a thick sheet of plastic, which is light in weight, yet strong enough to hold a load of several pounds. This load will come from the different activities of the prosthetic, such as grasping or carrying objects.

CHAPTER 4

SYSTEM REQUIREMENTS

4.1 HARDWARE REQUIREMENTS

PROCESSOR	:	Intel Core i7 8 th Gen
HARD DISK DRIVE	:	1TB
RAM	:	8 GB
GPU	:	NVIDIA GeForce GTX 1650

4.2 SOFTWARE REQUIREMENTS

OPERATING SYSTEM	:	Microsoft Windows 10
IDE	:	ARDUINO
LANGUAGES	:	Arduino Programming

4.3 DESCRIPTION OF THE COMPONENTS USED

The following are the tools used:

- EMG Sensor
- EMG Cables With Electrodes
- Arduino Board
- Servo Motor
- Prosthetic Arm

4.3.1 EMG Sensor

Electromyography (EMG) measures muscle response or electrical activity in response to a nerve's stimulation of the muscle. The test is used to help detect neuromuscular abnormalities. During the test, one or more small needles (also called electrodes) are inserted through the skin into the muscle. The electrical activity picked up by the electrodes is then displayed on an oscilloscope (a monitor that displays electrical activity in the form of waves). An audio-amplifier is used so the activity can be heard. EMG measures the

electrical activity of muscle during rest, slight contraction and forceful contraction. Muscle tissue does not normally produce electrical signals during rest.



Fig 4.3.1 Emg Sensor

4.3.2 Electrodes

EMG Electrodes provide a non - invasiv technique for measurement and detection of EMG Signal. The theory behind these electrodes is that they form a chemical equilibrium between the detecting surface and skin of the body through electrolytic conduction , so that current can flow into the electrodes.

These Electrodes are simple and very easy to implement. Application of Needle and fine wire Electrodes require strict medical supervision and certification. Surface EMG Electrodes require no such formalities. Surface EMG Electrodes have found their use in motor behavior study, neuromuscular recordings , sports medical evaluations and for subjects who object to needle insertions such as children. Apart from all this , Surface

EMG is being increasingly used to detect muscle activity in order to control device extensions to achieve prosthesis for physically disable and amputated populations.

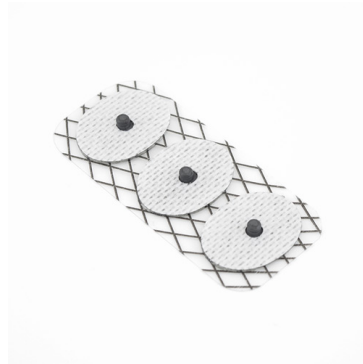


Fig 4.3.2 Electrodes

4.3.3 Arduino Board

Arduino is an open-source electronics platform based on easy-to-use hardware and software. Arduino Boards are able to read inputs - light on a sensor, a finger on a button, or a Twitter message - and turn it into an output - activating a motor, turning on an LED, publishing something online. the Arduino Programming Language (based on Wiring), and the Arduino Software (IDE), based on Processing .

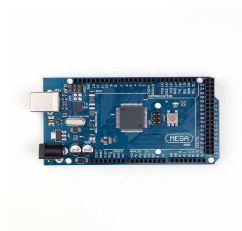


Fig 4.3.3 Arduino Mega

4.3.4 Servo Motor

Servo motors are great devices that can turn to a specified position.

Usually, they have a servo arm that can turn 180 degrees. Using the Arduino, we can tell a servo to go to a specified position and it will go there. As simple as that!

Servo motors were first used in the Remote Control (RC) world, usually to control the steering of RC cars or the flaps on a RC plane. With time, they found their uses in robotics, automation, and of course, the Arduino world.



Fig 4.3.4 Servo Motor

4.3.5 Prosthetic Arm

If you are missing an arm or leg, an artificial limb can sometimes replace it. The device, which is called a prosthesis, can help you to perform daily activities such as walking, eating, or dressing. Some artificial limbs let you function nearly as well as before. Prostheses are intended to restore the normal functions of the missing body

part. Prostheses can be created by hand or with computer Aided Design (CAD), a software interface that helps creators design and analyze the creation with computer-generated 2-D and 3- D Graphics as well as analysis and optimization tools.



Fig 4.3.5 Prosthetic Arm

CHAPTER 5

EMG Signal Processing

5.1 Introduction

This chapter discusses the mechanism of the muscle contractions and how to sense these activities. An explanation will be given for the way to process the signals that have been detected from the muscle to be readable by the controller, and finally, this chapter will show how to control the design and how to create an action dependent upon the incoming signal. There are three types of muscles in human bodies; skeletal muscles, which are connected to the skeleton and are responsible for body motions; smooth muscles, which are found in the walls of the human body's visceral organs except for the heart; and cardiac muscles, which are found in the heart's walls and are responsible for the heart's activities. This chapter focuses on the skeletal muscles which are responsible for human activities.

5.2 EMG Signal Detecting and Processing

With surface EMG, two electrodes should be used for each single muscle, and at least one electrode will be common between all muscles as shown in Figure 6.4 below. The first electrode (electrode one), should be connected to the middle of the muscle. Electrode two should be connected to the end of the muscle, and the last electrode will connect close to the bone as a ground (the ground could be common between many muscles). The reason for using two electrodes for the

muscle is to remove some of the noise signals. The differential amplifier will be used to subtract the signal that comes from electrode two from the signal which comes from electrode one to enhance the overall signal.

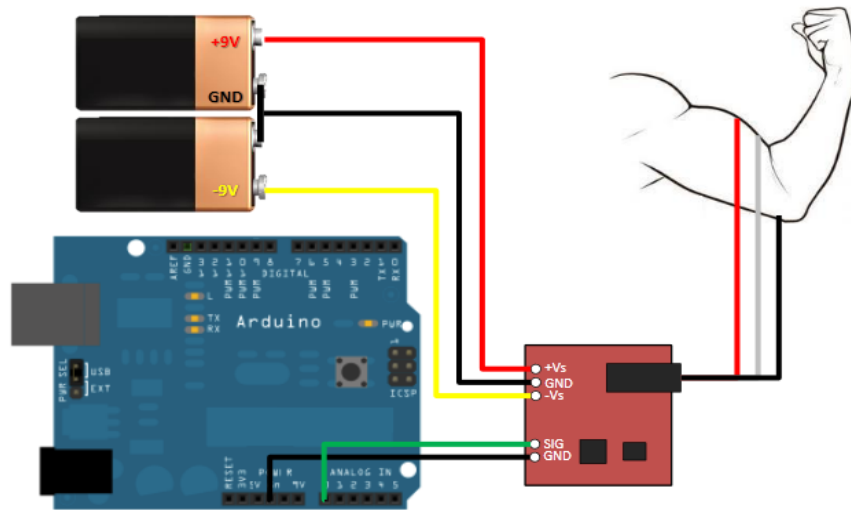


Fig 5.2 EMG Signal Detecting And Processing

Four things should be observed carefully during the detecting and processing of the EMG signal. The input impedance between the skin and the electrode can vary from several thousand ohms to several mega ohms. The input impedance should be as large as possible to prevent the distortion and attenuation of the detecting signal.

The preparation of the skin should emphasize that the skin needs to be clean and covered with gel to increase the electricity conduction.

CHAPTER 6

PROGRAMMING THE ARDUINO

The Arduino program will have two parts; dealing with the EMG signals and controlling servo motors. These signals are not regular signals, and they go up and down with different peak values. The incoming signal will be compared to the number which represents the signal strength that should have been received from training that muscle.

During the using of this prosthetic hand, the Arduino will continually check each analog pin if there is a signal or not, and to know how many fingers are activated in each single moment. In case there is a signal on one pin, and this signal is lying in the range of data that we got from the training, the Arduino will turn on the motor that corresponds to this muscle's signal. For example, if we suppose that the training of the pinky finger's signal during contraction was ranged between = 1.25 V ~ 2.0 V. This range will be saved in the Arduino program to use it for comparing with the data that comes from A0 when the amputee uses this prosthetic. The Arduino will continually check if the signal on pin A0 is lying in the range of 1.25 V ~ 2 V. If yes, that's means the pinky's muscle is in contraction, and the Arduino will control the motor that should actuate the pinky finger. When the signal goes down this value, that means the muscle became in rest again. Therefore, the Arduino will control the servo motor to open the pinky finger.

The same thing will happen if there is more than one finger activated at the same time. The Arduino will not turn on the motor of any finger unless the signal of that finger reaches the range of data that has been recorded in the training. And the signal will not be in that range of data unless the muscle was in contraction at that moment.

After reading the finger's incoming signal, the Arduino will start the next step, controlling the servo motors that actuates this finger. The controller will turn on the motor with 0 - 180 degrees to close or open the finger when there is a contraction signal, or the muscle will rest.

CHAPTER 7

CONCLUSION AND FUTURE WORK

7.1 CONCLUSION

This thesis project processes the design of a prosthetic hand controlled by the surface EMG signal. This cost-effective hand has 14 degree of freedom, and is made from aluminum sheet to be light and strong. The hand has five fingers, and each finger has three links except the thumb which has two links. Each finger is actuated separately by using servomotor which is controlled by Arduino. The Arduino controls the angle of the servomotor and the servomotor will pull a tie wrap to open or close the finger.

This prosthetic hand is controlled by the surface EMG signal which comes from the rest of the amputee's arm. The surface EMG signal should pass through many steps to be readable by the Arduino. The first step is detecting the EMG signal from the muscle by using three electrodes. Electrode one is connected to the top of the muscle, electrode two is connected to the end of muscle, and the third electrode is a reference electrode which is connected to the bone.

The reference electrode could be common between many muscles. That means if there is a circuit to test five muscles, it will need eleven electrodes. One will be common between all muscles and two electrodes for each muscle. After detecting a signal from the muscle, the next step will be to subtract the signal of electrode two from the signal of electrode one to remove the common noise. Then the signal should amplify several times to be large enough to pass through the next step. In

this pre-amplifying step, the signal will amplify for several times because it has noise and we don't want to amplify these noise. The next step is the filtering step where the signal should pass through the low pass filter then high pass filter to extract the EMG signal and remove almost all noise. The final step in EMG signal processing is the amplifying step where the signal will be amplified to be big enough to read by the Arduino to make control for the hand's fingers.

After designing the hand and getting the EMG signal, the project focused on getting a feedback signal from each finger. The feedback signal is the signal that tells the amputees if there is any object that is touching any finger. To do that, there is a force sensor connected to each fingertip which senses if there is any force applied on the finger. The way that the force sensor tells the amputee about this force, is by turning on the vibration motor which is connected to the amputee's arm, the speed of the vibration motor related with the amount of the force that is applied to the fingertip. This design has good functions that make it very good and useful for amputees. It is affordable, has light weight, easy to fix, good finger control, and very good EMG detecting. I hope this project will make a good difference in many amputees' lives.

7.2 FUTURE WORK AND RECOMMENDATIONS

After successfully designing and testing this project there are some recommendations for future work on this project. First of all, the feedback sensing ability should be improved. As mentioned before, each finger has a force sensor attached to the fingertip that can sense if there is something in touch with the finger. To improve the ability of sensing it is very important to increase the ability of the hand to sense in any part of the hand or fingers not just in fingertip. To do that, I recommended creating an artificial skin which covers all hands and has a good sensing ability and good friction to catch the object.

On the other hand, this design can tell the amputee if the object touches the finger hard or softly. But it cannot tell the amputee more properties about this object such as the temperature, if it is soft or not, if it is big or small, the shape of this object ...etc. Some of these properties are very important for blind people. To make this project very useful for blind people it is important to activate some feedback which tells the amputee about these properties.

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